



Question the Headline

Five lessons on questioning what you read, each built around a claim that was once printed as true and later fell apart. Students stand on a line to show how much they believe it, then take the evidence in four rounds, logging each card on their own evidence sheet and working out what the numbers show. They finish by writing the front page the paper should have printed. Each lesson needs a set of evidence cards per group and two sheets per student, and the lessons run in any order.

Before you start. Lesson 2 is about thalidomide and lesson 3 about smoking. A student may have a limb difference, or somebody at home who smokes, so keep both lessons on the claims and the evidence, show no photographs and never ask about home. Lessons 1 and 5 are about how food was sold, not about what anybody eats.

Print these

Page	What it is	How many
Page 2	Five questions for any headline, used in every lesson	One per group, or one on the wall
Page 3	The front page sheet, used in every lesson	One per student or pair
Pages 4 to 18	For each lesson: a teacher page with the hour and the answers, the evidence cards, and the evidence sheet	Cards: one set per group. Evidence sheet: one each

How it works

Step	What happens	Slides
1	The front page goes up and everybody stands on a line from 0 to 10 to show how much they believe it.	3 to 4
2	Teach the five questions, model the first one, then groups try the rest and fill in the top of their evidence sheets.	5 to 7
3	Four rounds of evidence. After each card, students log it on their sheet and move on the line. Card C comes with three questions to work out.	8 to 13
4	Each student or pair writes the front page the paper should have printed, then marks somebody else's.	14 to 15
5	The verdict, a question with no right answer, and two private questions on the evidence sheet.	16 to 18

Worth knowing

- Short of time? Skip the answers slide and the argument, and keep the rounds brisk.
- The cards can be cut up and handed out one per round, or printed whole as a sheet for each group if you would rather not cut. The front pages are recreated; the quotes and numbers on the cards are real.

Make sure you come back for Volume Two. Follow @thepastoralguide to hear when it lands.





Five questions for any headline

One per group on the table, or on the wall where everybody can see it. It is the same five in every lesson.

1



Who is saying it?

A company, a scientist, a newspaper, a government? Check the name.

2



Who gains if I believe it?

Somebody always does. Work out who, and what they get.

3



Who paid for it?

The research, the advert, the survey. Money does not make it false, but you should know.

4



What is the evidence?

A study, a number, a graph? Or only a slogan and a famous face?

5



Has it been tested, and by whom?

Tested on whom, for how long, and by somebody with nothing to gain?



The real front page

Write the front page the paper should have printed, on your own or with one partner. Write it for somebody who has not seen the cards.

WRITTEN BY

LESSON

The Daily Claim

THE CORRECTED EDITION

VOLUME ONE

WRITTEN FROM THE EVIDENCE

HEADLINE, TRUE TO THE EVIDENCE

ONE LINE UNDER IT: WHO MADE THE CLAIM, AND WHAT THEY STOOD TO GAIN

THE STORY, IN FOUR OR FIVE SENTENCES. USE AT LEAST TWO CARDS, NAME EACH ONE, AND PUT IN ONE NUMBER FROM YOUR SUMS.

CARDS I USED ☐ A ☐ B ☐ C ☐ D



Marked by somebody from another group. Tick what it passes, then write one change.

☐

True to the evidence

☐

Says where it came from

☐

Survives the five questions

ONE CHANGE





Who Paid for the Science?

The claim: "Sugar is not a problem for your heart. Fat is." This lesson is about who pays for research, and why readers need to be told.

The hour

TIME	WHAT HAPPENS	SLIDES
5 min	Put the front page up and read the headline and the first lines aloud.	3
4 min	Everybody stands on the line between 0 and 10. Hear two or three reasons.	4
5 min	Teach the five questions, then model the first one on the front page.	5 to 6
4 min	Groups try the other four on the front page. Everybody fills in the top of their evidence sheet.	7
8 min	Cards A and B, one round each. Everybody logs the card on their sheet, then moves on the line.	8 to 9
8 min	Card C, then the numbers: three questions worked out in groups, then the answers.	10 to 12
4 min	Card D, logged the same way, and a last move on the line.	13
10 min	Each student, or pair, writes the front page the paper should have printed.	14
5 min	Front pages swap, and each one is marked against the three checks.	15
4 min	The verdict, then a question with no right answer.	16 to 17
2 min	Two private questions, in the last box of the evidence sheet.	18

Crunch the numbers: the answers

- The fee was \$6,500 in 1965, about \$48,000 in 2016 money. Roughly how many times more is the 2016 figure?** About 7 times ($48,000 \div 6,500 = 7.4$).
- How many years passed between the review coming out (1967) and the letters being published (2016)?** 49 years.
- The journal started asking authors who paid them in 1984. How many years after the review was that, and why might it matter?** 17 years. Until then, readers had no way of knowing who had paid for a study.

The verdict, and what to watch for

The verdict is not that fat is harmless and sugar is the real villain. Current advice says to limit both. What was busted is the idea that the review was independent: the industry paid for it, saw it before publication and was pleased, and readers were never told. The question that catches it is Who paid for it?

Sources

- Kearns C. E., Schmidt L. A. and Glantz S. A. (2016). Sugar industry and coronary heart disease research. *JAMA Internal Medicine*, 176(11), 1680–1685.
- Bailey M. (2016). Sugar industry secretly paid for favorable Harvard research. *STAT News*, 12 September 2016.





The evidence: Who Paid for the Science?

One set per group, cut along the dashed lines. Hand out one card per round, in order.

EVIDENCE A

The authors

The review was written by nutrition scientists at Harvard, one of the most famous universities in the world. It came out in 1967 in the New England Journal of Medicine. The authors listed some of the money that paid for their work. The Sugar Research Foundation was not on the list.

STAT News, 12 September 2016

EVIDENCE B

The plan

In 1964, a sugar industry group wrote a plan to change “negative attitudes towards sugar”, partly by paying for research. In July 1965, one of its leaders, John Hickson, asked Harvard for a review of the studies that found “some special metabolic peril in sucrose”, which is sugar. The fee was \$6,500, about \$48,000 in today’s money.

STAT News and JAMA Internal Medicine, 2016

EVIDENCE C

The letter

The industry saw a draft before the review was sent to the journal. Hickson wrote back: “Let me assure you this is quite what we had in mind and we look forward to its appearance in print.” The review said cutting fat and cholesterol was the only diet change that could prevent heart disease.

Quoted in STAT News, 2016

EVIDENCE D

Fifty years later

In 2016, a researcher called Cristin Kearns published the letters, which she found in a Harvard library. Her co-author said the review “delayed the development of a scientific consensus on sugar–heart disease for decades.” Another Harvard scientist pointed out that, in the 1960s, the evidence against fat really was stronger.

Kearns, Schmidt and Glantz, JAMA Internal Medicine, 2016





Evidence sheet: Who Paid for the Science?

One each. Fill it in as the cards arrive. It is yours to keep, and nothing on it is read out.

NAME _____

Before the evidence

The claim: "Sugar is not a problem for your heart. Fat is."

WHERE I STAND ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 0 is definitely false, 10 is definitely true

WHAT WOULD I NEED TO KNOW BEFORE I BELIEVED IT?

The evidence

CARD	WHAT IT TELLS US, IN ONE SENTENCE	WHICH QUESTION IT ANSWERS	WHERE I STAND NOW
A The authors		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
B The plan		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
C The letter		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
D Fifty years later		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	

1 Who is saying it? · 2 Who gains if I believe it? · 3 Who paid for it? · 4 What is the evidence? · 5 Has it been tested, and by whom?

Crunch the numbers

- 1 The fee was \$6,500 in 1965, about \$48,000 in 2016 money. Roughly how many times more is the 2016 figure?

- 2 How many years passed between the review coming out (1967) and the letters being published (2016)?

- 3 The journal started asking authors who paid them in 1984. How many years after the review was that, and why might it matter?

After

WHERE DID I END UP, AND WHICH CARD MOVED ME MOST?

WHICH OF THE FIVE QUESTIONS WILL I ASK FIRST NEXT TIME?





Complete Safety

The claim: "If a medicine is sold by a big company and prescribed by doctors, it must have been tested properly." This lesson is about how medicines are tested before they are sold, and the official who kept asking for better evidence.

The hour

TIME	WHAT HAPPENS	SLIDES
5 min	Put the front page up and read the headline and the first lines aloud.	3
4 min	Everybody stands on the line between 0 and 10. Hear two or three reasons.	4
5 min	Teach the five questions, then model the first one on the front page.	5 to 6
4 min	Groups try the other four on the front page. Everybody fills in the top of their evidence sheet.	7
8 min	Cards A and B, one round each. Everybody logs the card on their sheet, then moves on the line.	8 to 9
8 min	Card C, then the numbers: three questions worked out in groups, then the answers.	10 to 12
4 min	Card D, logged the same way, and a last move on the line.	13
10 min	Each student, or pair, writes the front page the paper should have printed.	14
5 min	Front pages swap, and each one is marked against the three checks.	15
4 min	The verdict, then a question with no right answer.	16 to 17
2 min	Two private questions, in the last box of the evidence sheet.	18

Crunch the numbers: the answers

- 1. About how many times more babies were harmed in the UK (about 2,000) than in the US (17)?** About 118 times ($2,000 \div 17 = 117.6$).
- 2. Distaval was on sale in the UK from 1958 to 1961. The advert said it had been prescribed for "nearly three years". Does that fit?** Yes: 1958 to 1961 is about three years, so the advert ran near the end, when the harm was already happening.
- 3. Card B says what was tested and what was not. What should have been tested before it was sold for morning sickness?** What the drug did to a baby before birth.

The verdict, and what to watch for

Some students may have a disability or a limb difference, or a family member who does. Keep the lesson on the company's claims and the evidence, and do not show photographs. The US bar is small because the drug was never approved there, not because Americans were luckier. The question that catches it is Has it been tested, and by whom?

Sources

- Wikipedia, Thalidomide and Thalidomide scandal, accessed September 2026.
- Phillips S. (2020). How a courageous physician-scientist saved the U.S. from a birth-defects catastrophe. UChicago Medicine.
- McNeill L. (2017). The woman who stood between America and a generation of thalidomide babies. Smithsonian Magazine.





The evidence: Complete Safety

One set per group, cut along the dashed lines. Hand out one card per round, in order.

EVIDENCE A

The advert

An advert for Distaval told doctors: “Distaval can be given with complete safety to pregnant women and nursing mothers without adverse effect on mother or child.” It went on: “Outstandingly safe Distaval has been prescribed for nearly three years in this country.”

Quoted in Wikipedia, Thalidomide

EVIDENCE B

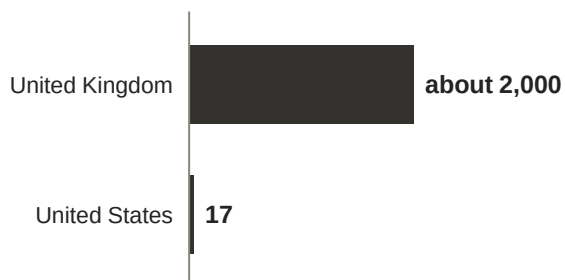
The testing

At the time, new medicines were tested on rats and mice to check they were not poisonous. Nobody had to test what a medicine did to a baby before birth, and thalidomide was never tested in pregnancy before it went on sale.

Wikipedia, Thalidomide; Science Museum

EVIDENCE C

The numbers



Both countries had the drug. In the UK it was on sale from 1958 to 1961. In the US it was only handed out in trials, to doctors testing it.

Wikipedia, Thalidomide scandal; UChicago Medicine

EVIDENCE D

The official who said no

In September 1960, Frances Kelsey at the US medicines regulator was asked to approve thalidomide. She thought the evidence was thin. “It was just too positive; this couldn’t be the perfect drug with no risk,” she said. She asked for proof it was safe for an unborn baby, again and again, and never approved it.

UChicago Medicine, 2020; Smithsonian, 2017





Evidence sheet: Complete Safety

One each. Fill it in as the cards arrive. It is yours to keep, and nothing on it is read out.

NAME _____

Before the evidence

The claim: "If a medicine is sold by a big company and prescribed by doctors, it must have been tested properly."

WHERE I STAND ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 0 is definitely false, 10 is definitely true

WHAT WOULD I NEED TO KNOW BEFORE I BELIEVED IT?

The evidence

CARD	WHAT IT TELLS US, IN ONE SENTENCE	WHICH QUESTION IT ANSWERS	WHERE I STAND NOW
A The advert		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
B The testing		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
C The numbers		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
D The official who said no		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	

1 Who is saying it? · 2 Who gains if I believe it? · 3 Who paid for it? · 4 What is the evidence? · 5 Has it been tested, and by whom?

Crunch the numbers

1 About how many times more babies were harmed in the UK (about 2,000) than in the US (17)?

2 Distaval was on sale in the UK from 1958 to 1961. The advert said it had been prescribed for "nearly three years". Does that fit?

3 Card B says what was tested and what was not. What should have been tested before it was sold for morning sickness?

After

WHERE DID I END UP, AND WHICH CARD MOVED ME MOST?

WHICH OF THE FIVE QUESTIONS WILL I ASK FIRST NEXT TIME?





Doctors' Orders

The claim: "If doctors smoke them, cigarettes can't be that bad for you." This lesson is about how adverts used doctors to sell cigarettes, and the study that tested the claim.

The hour

TIME	WHAT HAPPENS	SLIDES
5 min	Put the front page up and read the headline and the first lines aloud.	3
4 min	Everybody stands on the line between 0 and 10. Hear two or three reasons.	4
5 min	Teach the five questions, then model the first one on the front page.	5 to 6
4 min	Groups try the other four on the front page. Everybody fills in the top of their evidence sheet.	7
8 min	Cards A and B, one round each. Everybody logs the card on their sheet, then moves on the line.	8 to 9
8 min	Card C, then the numbers: three questions worked out in groups, then the answers.	10 to 12
4 min	Card D, logged the same way, and a last move on the line.	13
10 min	Each student, or pair, writes the front page the paper should have printed.	14
5 min	Front pages swap, and each one is marked against the three checks.	15
4 min	The verdict, then a question with no right answer.	16 to 17
2 min	Two private questions, in the last box of the evidence sheet.	18

Crunch the numbers: the answers

- 1. How many times higher was the lung cancer death rate for doctors who smoked 25 or more a day (2.27) than for non-smokers (0.07)?** About 32 times ($2.27 \div 0.07 = 32.4$).
- 2. What happens to the death rate every time the number of cigarettes goes up a band?** It rises every time: 0.07, then 0.57, then 1.39, then 2.27.
- 3. The rates are per 1,000 doctors a year. Out of 1,000 doctors who smoked 25 or more a day, about how many would die of lung cancer over ten years?** About 23 (2.27×10). For non-smokers it would be fewer than 1.

The verdict, and what to watch for

Some students will have parents or carers who smoke. The lesson is about how the adverts worked, not about the people who smoke, and nobody should be asked about home. The busted claim is that doctors smoking a brand made it safe; a survey with no method is not evidence. The question that catches it is What is the evidence?

Sources

- Doll R. and Hill A. B. (1950). Smoking and carcinoma of the lung. British Medical Journal, 2(4682), 739-748.
- Doll R. and Hill A. B. (1964). Mortality in relation to smoking: ten years' observations of British doctors. British Medical Journal, 1, 1399-1410. Rates as set out in the CDC's case study on this work.
- Center for the Study of Tobacco and Society, University of Alabama: More Doctors Smoke Camels.





The evidence: Doctors' Orders

One set per group, cut along the dashed lines. Hand out one card per round, in order.

EVIDENCE A

The advert

In 1946, R.J. Reynolds began a campaign with the slogan “More doctors smoke Camels than any other cigarette.” It claimed to be based on a survey of 113,597 doctors. The adverts ran in magazines, on the radio and in medical journals for several years.

Center for the Study of Tobacco and Society, University of Alabama

EVIDENCE B

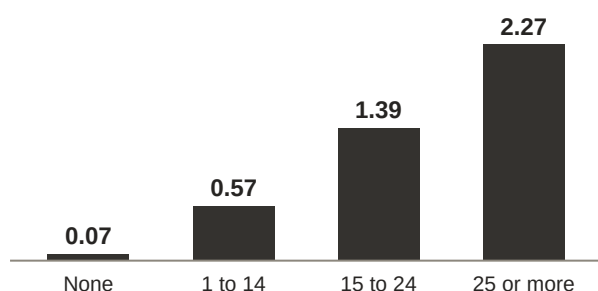
The survey

The company never explained how the survey was done. One account says doctors were given a free carton of Camels and then asked what brand they smoked. In 1930, another company sent doctors free cartons of its own cigarettes and then advertised that 20,679 of them said its cigarettes were “less irritating”.

History.com, 2018; CSTS, University of Alabama

EVIDENCE C

The doctors' own study



Deaths from lung cancer per 1,000 British doctors a year, by cigarettes smoked a day, 1951 to 1961.

Doll and Hill, British Medical Journal, 1964

EVIDENCE D

The scientists

In 1950, Richard Doll and Austin Bradford Hill studied patients in twenty London hospitals. They wrote that the risk of lung cancer “may be 50 times as great among those who smoke 25 or more cigarettes a day as among non-smokers.” Doll had been a smoker. He gave up.

Doll and Hill, British Medical Journal, 1950





Evidence sheet: Doctors' Orders

One each. Fill it in as the cards arrive. It is yours to keep, and nothing on it is read out.

NAME _____

Before the evidence

The claim: "If doctors smoke them, cigarettes can't be that bad for you."

WHERE I STAND ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 0 is definitely false, 10 is definitely true

WHAT WOULD I NEED TO KNOW BEFORE I BELIEVED IT?

The evidence

CARD	WHAT IT TELLS US, IN ONE SENTENCE	WHICH QUESTION IT ANSWERS	WHERE I STAND NOW
A The advert		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
B The survey		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
C The doctors' own study		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	
D The scientists		☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5	

1 Who is saying it? · 2 Who gains if I believe it? · 3 Who paid for it? · 4 What is the evidence? · 5 Has it been tested, and by whom?

Crunch the numbers

1 How many times higher was the lung cancer death rate for doctors who smoked 25 or more a day (2.27) than for non-smokers (0.07)?

2 What happens to the death rate every time the number of cigarettes goes up a band?

3 The rates are per 1,000 doctors a year. Out of 1,000 doctors who smoked 25 or more a day, about how many would die of lung cancer over ten years?

After

WHERE DID I END UP, AND WHICH CARD MOVED ME MOST?

WHICH OF THE FIVE QUESTIONS WILL I ASK FIRST NEXT TIME?





No Real Evidence

The claim: "If nobody has proved something is harmful, it must be safe." This lesson is about the difference between no evidence of harm and evidence that something is safe.

The hour

TIME	WHAT HAPPENS	SLIDES
5 min	Put the front page up and read the headline and the first lines aloud.	3
4 min	Everybody stands on the line between 0 and 10. Hear two or three reasons.	4
5 min	Teach the five questions, then model the first one on the front page.	5 to 6
4 min	Groups try the other four on the front page. Everybody fills in the top of their evidence sheet.	7
8 min	Cards A and B, one round each. Everybody logs the card on their sheet, then moves on the line.	8 to 9
8 min	Card C, then the numbers: three questions worked out in groups, then the answers.	10 to 12
4 min	Card D, logged the same way, and a last move on the line.	13
10 min	Each student, or pair, writes the front page the paper should have printed.	14
5 min	Front pages swap, and each one is marked against the three checks.	15
4 min	The verdict, then a question with no right answer.	16 to 17
2 min	Two private questions, in the last box of the evidence sheet.	18

Crunch the numbers: the answers

- By roughly what percentage did the lead in young children's blood fall from 1976-80 (15.2) to 2017-18 (0.7)?** About 95 per cent ($(15.2 - 0.7) \div 15.2 = 0.954$).
- Between which two surveys on the graph was the biggest fall?** Between 1976-80 and 1999-2000, from 15.2 to 2.2, which is when lead came out of petrol.
- Name one other change that could explain some of the fall.** Lead paint and lead solder were banned in the same years.

The verdict, and what to watch for

The graph shows lead in blood falling as lead came out of petrol, but lead paint and lead solder were banned in the same years, and a sharp class will say so. That is the right answer, not a problem: the US Environmental Protection Agency says the biggest fall followed the removal of lead from petrol. The question that catches the 1926 claim is Has it been tested, and by whom?

Sources

- Egan K. B. and others (2021). Blood lead levels in U.S. children ages 1-11 years, 1976-2016. *Environmental Health Perspectives*, 129(3).
- US Environmental Protection Agency, Report on the Environment: Blood Lead (2023) and America's Children and the Environment: Lead (2026).
- UNEP (2021). Era of leaded petrol over. Press release, 30 August 2021.
- Wikipedia, Tetraethyllead, Robert A. Kehoe and Clair Cameron Patterson, accessed September 2026.
- Ericson J. E., Shirahata H. and Patterson C. C. (1979). Skeletal concentrations of lead in ancient Peruvians. *New England Journal of Medicine*, 300, 946-951.





The evidence: No Real Evidence

One set per group, cut along the dashed lines. Hand out one card per round, in order.

EVIDENCE A

The deaths

In 1924, five workers died and many more were badly hurt at Standard Oil refineries in New Jersey, where the lead additive was made. Newspapers called it “loony gas”. Sales stopped for a year. In 1926, a committee said there was “no real evidence” of harm to the public, and sales started again.

Wikipedia, Tetraethyllead

EVIDENCE B

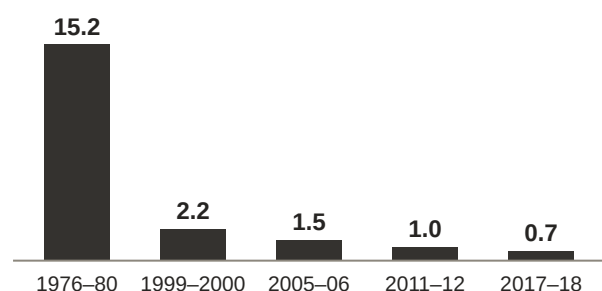
The expert

For forty years the leading expert on lead was Robert Kehoe. He said the lead found in people was normal and that low levels did no harm. From 1925 he was the chief medical adviser to the Ethyl Corporation, which sold the lead additive, and his laboratory was paid for by General Motors, DuPont and Ethyl.

Wikipedia, Robert A. Kehoe; University of Cincinnati Libraries

EVIDENCE C

The numbers



Average lead in the blood of US children aged 1 to 5, in micrograms per decilitre. The US took lead out of petrol between 1975 and 1996.

Egan and others, 2021; US EPA

EVIDENCE D

The scientist

Clair Patterson set out to measure the age of the Earth and kept finding lead everywhere. He showed that ice laid down before leaded petrol held very little lead, and his team found that modern bones held hundreds of times more lead than bones from centuries ago. He argued that “normal” was not the same as “natural”.

Wikipedia, Clair Cameron Patterson





Evidence sheet: No Real Evidence

One each. Fill it in as the cards arrive. It is yours to keep, and nothing on it is read out.

NAME _____

Before the evidence

The claim: "If nobody has proved something is harmful, it must be safe."

WHERE I STAND 0 1 2 3 4 5 6 7 8 9 10 0 is definitely false, 10 is definitely true

WHAT WOULD I NEED TO KNOW BEFORE I BELIEVED IT?

The evidence

CARD	WHAT IT TELLS US, IN ONE SENTENCE	WHICH QUESTION IT ANSWERS	WHERE I STAND NOW
A The deaths		1 2 3 4 5	
B The expert		1 2 3 4 5	
C The numbers		1 2 3 4 5	
D The scientist		1 2 3 4 5	

1 Who is saying it? · 2 Who gains if I believe it? · 3 Who paid for it? · 4 What is the evidence? · 5 Has it been tested, and by whom?

Crunch the numbers

1 By roughly what percentage did the lead in young children's blood fall from 1976-80 (15.2) to 2017-18 (0.7)?

2 Between which two surveys on the graph was the biggest fall?

3 Name one other change that could explain some of the fall.

After

WHERE DID I END UP, AND WHICH CARD MOVED ME MOST?

WHICH OF THE FIVE QUESTIONS WILL I ASK FIRST NEXT TIME?





It's All Vegetable

The claim: "A new food made in a factory by scientists is probably better for you than the old one." This lesson is about how a new food was sold, and what an experiment showed about it later.

The hour

TIME	WHAT HAPPENS	SLIDES
5 min	Put the front page up and read the headline and the first lines aloud.	3
4 min	Everybody stands on the line between 0 and 10. Hear two or three reasons.	4
5 min	Teach the five questions, then model the first one on the front page.	5 to 6
4 min	Groups try the other four on the front page. Everybody fills in the top of their evidence sheet.	7
8 min	Cards A and B, one round each. Everybody logs the card on their sheet, then moves on the line.	8 to 9
8 min	Card C, then the numbers: three questions worked out in groups, then the answers.	10 to 12
4 min	Card D, logged the same way, and a last move on the line.	13
10 min	Each student, or pair, writes the front page the paper should have printed.	14
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4 min	The verdict, then a question with no right answer.	16 to 17
2 min	Two private questions, in the last box of the evidence sheet.	18

Crunch the numbers: the answers

1. **Which fat raised the "bad" LDL cholesterol more, trans fat (+0.37) or saturated fat (+0.47)?** Saturated fat, by 0.10.
2. **Which fat lowered the "good" HDL cholesterol, and by how much?** Trans fat, by 0.17. Saturated fat did not change it.
3. **Add up how far each fat moved the two numbers in the wrong direction. Which comes out worse?** Trans fat: $0.37 + 0.17 = 0.54$. Saturated fat: $0.47 + 0 = 0.47$. Trans fat is worse, but neither is harmless.

The verdict, and what to watch for

This lesson is about how a product was sold and what the evidence later showed, not about what anybody eats. Keep it off good and bad foods and off anybody's lunch. The data does not make butter healthy: saturated fat raised the bad cholesterol slightly more than trans fat did. What was busted is the claim that the factory fat was the healthier choice. The question that catches it is Who gains if I believe it?

Sources

- Mensink R. P. and Katan M. B. (1990). Effect of dietary trans fatty acids on high-density and low-density lipoprotein cholesterol levels in healthy subjects. *New England Journal of Medicine*, 323(7), 439-445.
- World Health Organization (2024). Trans fat: fact sheet.
- US Food and Drug Administration. Final determination regarding partially hydrogenated oils (2015).
- NPR (2012). The forgotten, fascinating saga of Crisco. Wikipedia, Crisco, accessed September 2026.





The evidence: It's All Vegetable

One set per group, cut along the dashed lines. Hand out one card per round, in order.

EVIDENCE A

The company

Procter & Gamble made soap and candles. Crisco was made from cottonseed oil, which comes from the seeds left over when cotton is grown. Around 1907, with the chemist Edwin Kayser, the company found a way to turn the oil into a solid fat. It sold that fat as food from June 1911, and named it Crisco, from “crystallized cottonseed oil”.

Wikipedia, Crisco; University of Guelph

EVIDENCE B

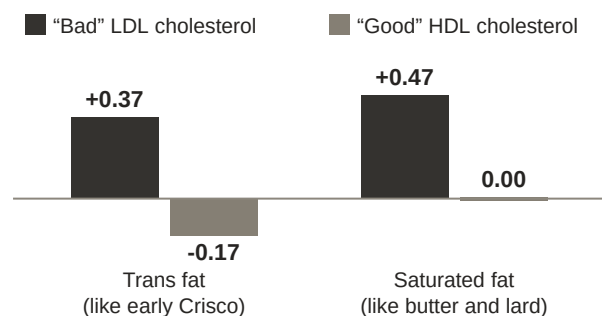
The adverts

Crisco was advertised as healthier and more cost-effective than butter and other fats. One slogan was “It’s all vegetable! It’s digestible!” The company gave away free cookbooks in which every recipe used Crisco, and its adverts asked shoppers to trust the brand rather than worry about what was in it.

NPR, 2012; Wikipedia, Crisco; University of Guelph

EVIDENCE C

The experiment



59 healthy volunteers ate three diets for three weeks each, and only one fat changed. Changes in mmol per litre, compared with a diet high in olive-oil type fat.

Mensink and Katan, New England Journal of Medicine, 1990

EVIDENCE D

What happened

Turning liquid oil solid makes trans fat. The World Health Organization now links up to 278,000 deaths a year to trans fats made in factories. The US ruled them unsafe in 2015 and gave food companies until 2018 to remove them. Crisco changed its recipe in 2007.

WHO trans fat fact sheet, 2024; US FDA; Wikipedia, Crisco





Evidence sheet: It's All Vegetable

One each. Fill it in as the cards arrive. It is yours to keep, and nothing on it is read out.

NAME _____

Before the evidence

The claim: "A new food made in a factory by scientists is probably better for you than the old one."

WHERE I STAND 0 1 2 3 4 5 6 7 8 9 10 0 is definitely false, 10 is definitely true

WHAT WOULD I NEED TO KNOW BEFORE I BELIEVED IT?

The evidence

CARD	WHAT IT TELLS US, IN ONE SENTENCE	WHICH QUESTION IT ANSWERS	WHERE I STAND NOW
A The company		1 2 3 4 5	
B The adverts		1 2 3 4 5	
C The experiment		1 2 3 4 5	
D What happened		1 2 3 4 5	

1 Who is saying it? · 2 Who gains if I believe it? · 3 Who paid for it? · 4 What is the evidence? · 5 Has it been tested, and by whom?

Crunch the numbers

1 Which fat raised the "bad" LDL cholesterol more, trans fat (+0.37) or saturated fat (+0.47)?

2 Which fat lowered the "good" HDL cholesterol, and by how much?

3 Add up how far each fat moved the two numbers in the wrong direction. Which comes out worse?

After

WHERE DID I END UP, AND WHICH CARD MOVED ME MOST?

WHICH OF THE FIVE QUESTIONS WILL I ASK FIRST NEXT TIME?

